Governing the DAO Treasury: Token-Weighted vs. Square-Root Voting under Whale Concentration and Sybil Splitting

Zimu Zhou | Team FP11 (solo) | Duke Kunshan University
Course instructor: Professor Luyao Zhang

昆山杜克大学 DUKE KUNSHAN UNIVERSITY

01 Question & model

Research question & verified gap

Shared question: When does SRV improve welfare and non-whale representation over TWV, and when does wallet splitting remove the gains?
Connected questions: Economics evaluates incentives and welfare. Computation tests wallet splitting at fixed concentration. Behavior asks whether fairness and Sybil awareness change rule preferences.
Verified gap: Fritsch et al. [4] measure actual DAO voting concentration. QV theory [2] prices votes, while Sybil research [3,5] already establishes identity risks. Our contribution is a bounded, reproducible comparison of these established mechanisms.

Voting model & foundations

Players: Eight actors vote A, B or C for one fixed grant. The whale owns c = 0.60. Seven holders each own 0.40/7. Whale wallets vote as a bloc.
Utility and values: Values v_{ij} are iid Uniform[0,1). Winner j^* gives $u_i = v_{ij}{}^*$. The simulator observes all values to evaluate welfare.
Timing and behavior: Set the rule and wallet count m, then each actor votes sincerely for their highest-valued proposal. This is a one-shot voting baseline; strategic deviations are not simulated.
Scoring: TWV uses $w_i = s_i$. SRV uses $w_i = \sqrt{s_i}$. Highest total score wins, with A/B/C tie priority. This is square-root weighting, not full QV.

Three complementary lenses

Game theory: Splitting c into m wallets raises SRV whale weight from √c to √(mc); TWV stays c. At c = 0.6, m > 4.67 gives the whale a voting majority under SRV. The simulation fixes m.
Social choice: Holder welfare $W(j) = \Sigma_i v_{ij}$ counts actors equally. Stake welfare $\Sigma_i s_i v_{ij}$ weights ownership. Better representation can reduce stake efficiency.
Mechanism design: Hold values fixed across TWV (m = 1), SRV (m = 1) and split SRV (m = 8).

Week 5: treasury allocation comparison

Resource and reports: One indivisible grant goes to one of three proposals. Holders report ballots based on individual valuations, not monetary bids.
Rules and outcomes: The highest score receives the grant. Voting has zero payments and zero DAO revenue. Utility is the value of the selected proposal.
Boundary: This is collective allocation. It has no seller, auction price, truthful-bidding theorem or revenue objective.

02 Design & evidence

Integrated research design

Economic model	Computational evidence	Behavioral evidence
Unequal token holdings TWV versus SRV Holder/stake welfare	60% whale; 3 voting conditions 1,000 shared value profiles Sincere ballots; seed 206	Browser self-play Rule preference and fairness Sybil-salience reflection
Prediction: less power for one unsplit whale.	Test: fixed stake, one versus eight whale wallets.	Revision: separate fairness from attack resistance.

Figure 1. One shared question links splitting incentives, reproducible voting outcomes and a bounded behavioral reflection. Cost and salience experiments remain planned.

Key message at 60% whale ownership

In this benchmark, SRV improves representation at m = 1. Eight-wallet splitting removes that advantage.

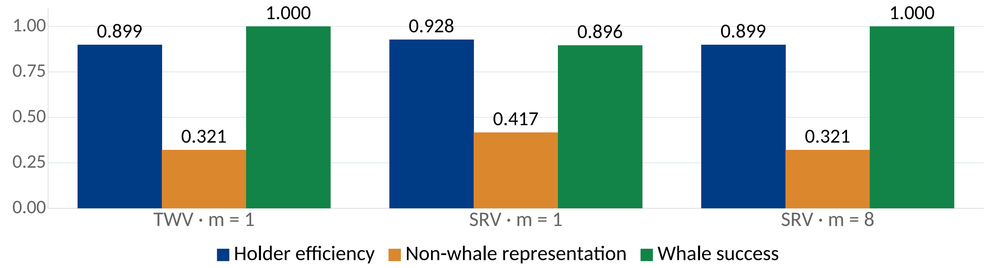


Figure 2. Sincere-voting means, seed 206, 1,000 shared profiles per condition. Probabilities and efficiency ratios use the same 0–1 scale.

Outcome (sincere voting)	TWV, m=1	SRV, m=1	SRV, m=8
Holder efficiency	0.899065	0.927781	0.899065
Stake-weighted efficiency	0.991097	0.977076	0.991097
Whale-preferred winner	100.0%	89.6%	100.0%
Non-whale representation	32.1%	41.7%	32.1%
Payments / DAO revenue	0 / 0	0 / 0	0 / 0

Metrics: Efficiency divides realized welfare by its feasible maximum. Representation is the probability that the winner maximizes the seven non-whales' summed values. It is not a headcount-majority measure.
Whale success is the fraction where the whale's favorite wins. Welfare counts the whale once, regardless of wallet count.

Reproducibility & verification

Python random.Random(206); c = 0.60; 1,000 shared profiles × 3 conditions = 3,000 rows. Committed CSV/JSON outputs agree with the executed notebook. Reruns are deterministic; TWV and split SRV match profile by profile.

03 Impact & dialogue

Practical case: Uniswap governance

Context: UNI holders govern treasury spending [6]. We examine requirements for an SRV grant-voting pilot, without estimating actual Uniswap outcomes.
Pilot test: Hold proposals fixed. Require representation gains to survive plausible splitting attacks and disclose identity safeguards' access costs.
Risks: Identity checks may exclude legitimate users and impose privacy and implementation costs.
Correction: Publish results, allow wallet-classification appeals, rerun disputed cases, and revise or suspend a pilot whose identity assumptions fail.

Behavior & interdisciplinary integration

Self-play, n = 1: The author voted A twice. TWV selected C and SRV selected A. After considering Sybil splitting, preference switched from SRV to TWV.
Reflection: Fairness mattered, yet the author would split a 60% whale stake into eight costless wallets. Fairness can conflict with incentives.
Alternative explanation: Personal outcomes may drive preference. One self-play record establishes neither causality nor a population pattern.
Integration: GitHub/Colab quantify synthetic voting outcomes. HF uses separate self-play scenarios to record fairness, rule preference and splitting intentions. These are distinct evidence sources.
Next test: Fix values and outcomes. Vary Sybil salience and splitting cost. Compare preferences and splitting intentions.

Limitations & next computational test

Scope: Synthetic iid values, sincere ballots and a 60% whale. Only the whale splits; splitting is exogenous, costless and coordinated.
Evidence boundary: Three conditions and one seed-206 sample. No equilibrium, uncertainty interval, real-DAO outcome or causal behavioral effect is established.
Next test: Add strategic voting and test Nash equilibria [1]. Vary concentration, splitting costs, identity constraints, seeds and valuation distributions.

Contribution, leadership & dialogue

Value: Academic stress tests, an auditable industry pilot, and accessible education with a public challenge route. SDG 4 learning impact is untested.
2056 aspiration: Accountable governance under unequal ownership and manipulable identities.
Question: Which safeguards limit splitting without excluding users?

Acknowledgments & responsibility

Zimu Zhou completed the model, code, self-play, paper and poster as a solo team. Professor Luyao Zhang provided course guidance. The poster adapts the supplied course template.
Tools: Python standard library, GitHub, Colab and Hugging Face. ChatGPT/Codex assisted drafting, debugging and layout. The author retains responsibility for assumptions, checks and claims.

Selected references

[1] Nash, J. F. 1950. "Equilibrium Points in n-Person Games." PNAS 36(1): 48–49. https://doi.org/10.1073/pnas.36.1.48
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[4] Fritsch, R., M. Müller, and R. Wattenhofer. 2024. "Analyzing Voting Power in Decentralized Governance: Who Controls DAOs?" Blockchain: Research and Applications 5(3): 100208. https://doi.org/10.1016/j.bcra.2024.100208
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[6] Uniswap. "Governance Overview." Accessed 27 Sep 2026. https://developers.uniswap.org/docs/ecosystem/governance/overview

Open materials & exact version

Hugging Face: https://huggingface.co/spaces/dku-comsci-econ206-2026/DAO_Treasury
GitHub: https://github.com/dku-comsci-econ206-Autumn2026/PS2_Zimu
Commit: 4e96644c9962f7d1441e7d6c36063d4b2facf3de
Colab: https://colab.research.google.com/github/dku-comsci-econ206-Autumn2026/PS2_Zimu/blob/main/dao_treasury_simulation.ipynb

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